RAG-Based Image Captioning Using Retrieval-Augmented Generation

Phase 1 Report: Dataset Exploration, Preprocessing and Retrieval Foundation

Group Project – Group 10
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Abstract

This report presents Phase 1 of our RAG-Based Image Captioning project using the MS COCO dataset. The main objective of this phase is to prepare, explore, and validate the dataset before implementing the assigned RAG technique. The work includes dataset splitting, statistical exploration, image preprocessing, object extraction using YOLOv8, image and text embedding generation using CLIP, and vector database construction using FAISS. The resulting retrieval foundation will be used in the next phase to investigate RAG Reranking techniques, specifically Cross-Encoder reranking.

1 Introduction

Image captioning is the task of generating natural-language descriptions of images. It requires understanding visual content and connecting it with meaningful textual information. Retrieval-Augmented Generation (RAG) can improve this process by retrieving relevant visual and textual information from a knowledge base before generating a caption. This project investigates RAG techniques for image captioning using the MS COCO dataset. The first phase focuses on preparing and analyzing the dataset, extracting visual information, generating embeddings, and constructing the initial retrieval system.

2 Dataset Description and Splitting

2.1 MS COCO Dataset

We use the MS COCO dataset. It contains images of everyday scenes, object annotations, and human-written captions. The dataset provides two main components: images with object annotations, and reference captions describing image content.

2.2 Dataset Splitting

To prepare the data for retrieval and evaluation, we created three separate splits:

Split	Source	Number of Images
Knowledge Base	COCO train2017	10,000
Validation	COCO train2017	100
Test	COCO val2017	1,000
Total		**11,100**

The Knowledge Base contains the images and reference captions used to construct the retrieval database. Validation is kept separate, while Test is reserved for held-out evaluation.

2.3 Caption Distribution

Split	Images	Captions
Knowledge Base	10,000	50,028
Validation	100	500
Test	1,000	5,002
Total	**11,100**	**55,530**

Most images have five reference captions. A small number have additional captions in the original COCO annotations.

3 Statistical Exploration of the Dataset

Statistical exploration was performed to understand the visual and linguistic characteristics of the Knowledge Base.

3.1 Object Category Distribution

We analyzed the original COCO object annotations for the 10,000 Knowledge Base images. The most frequent categories include:

Object Category	Number of Instances
Person	21,825
Car	3,784
Chair	3,163
Book	2,034
Bottle	1,847

Person is the most frequent category, showing that human-centered scenes are common in the selected dataset.

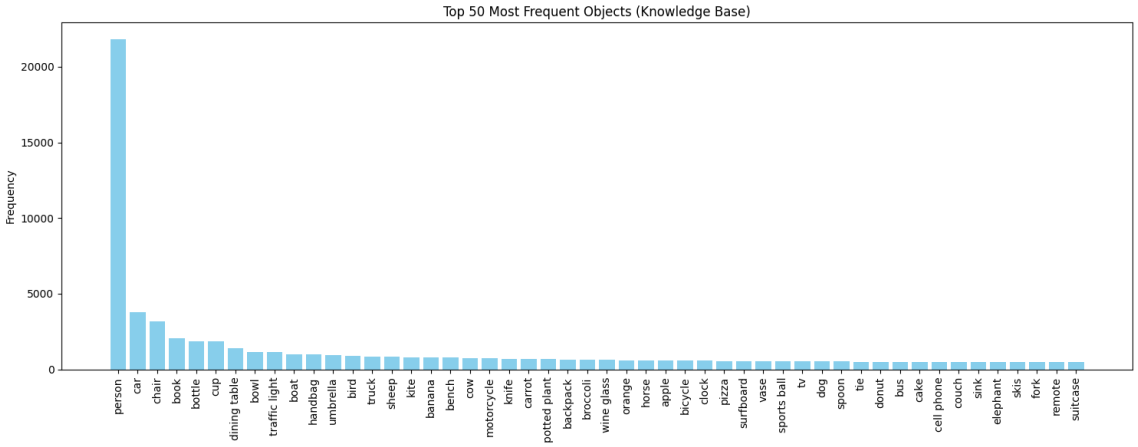


Figure 1: Top 50 most frequent object categories in the Knowledge Base based on COCO annotations.

3.2 Objects per Image

We analyzed the number of annotated objects in each image. The average number of objects per image is approximately 7.24.

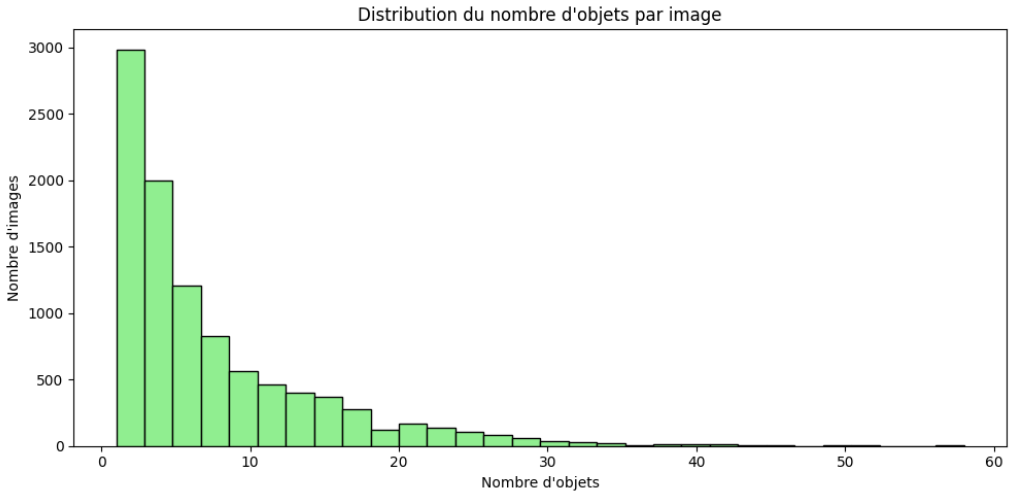


Figure 2: Distribution of the number of annotated objects per image in the Knowledge Base.

3.3 Caption Length Analysis

We analyzed the reference captions to understand their linguistic characteristics. The main statistics are: average caption length 10.46 words; median 10 words; minimum 7 words; maximum 46 words. The unique vocabulary contains 9,589 words across the Knowledge Base.

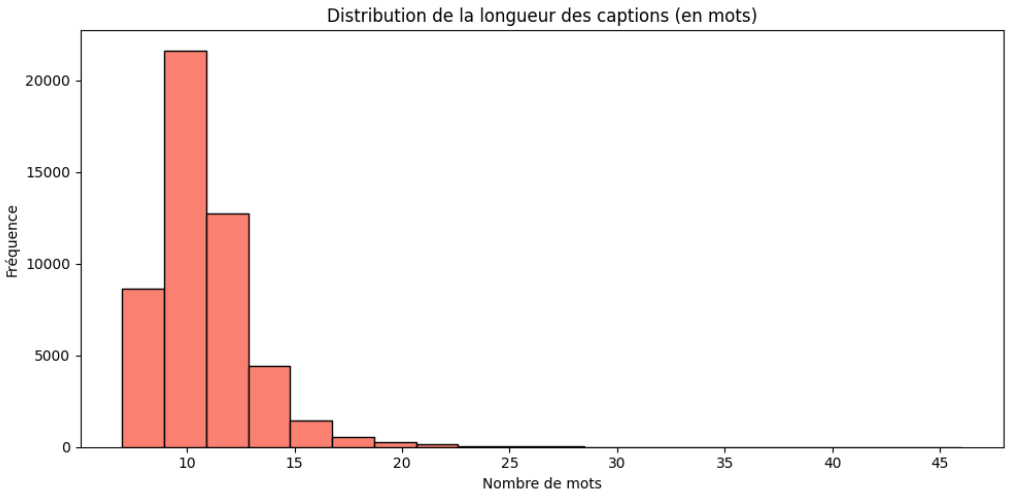


Figure 3: Distribution of caption length in words in the Knowledge Base.

3.4 Scene Context Analysis

We also performed an approximate indoor/outdoor analysis based on selected object categories. The estimated distribution was Indoor: 37.6% and Outdoor: 62.4%.

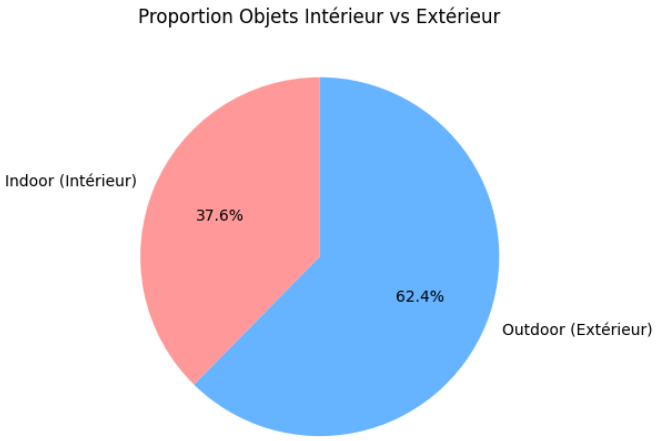


Figure 4: Approximate indoor/outdoor distribution inferred from selected object categories.

4 Data Audit and Preprocessing

4.1 Data Quality Audit

A data audit was conducted to verify the consistency and integrity of the selected dataset. The audit results confirmed that no images were missing, all images had captions, and no

corrupted images were detected.

4.2 Image Preprocessing

Images are opened using the Python PIL library and converted to RGB format during preprocessing. The CLIPProcessor subsequently performs the required image resizing, cropping, and pixel normalization during embedding generation.

4.3 Object Extraction Using YOLOv8

To extract visual information directly from images, we used the pretrained YOLOv8n object detection model with a confidence threshold of 0.25.

4.3.1 Knowledge Base Results

YOLOv8 processed 10,000 images and detected 53,872 objects. The average number of detections was 5.38 per image.

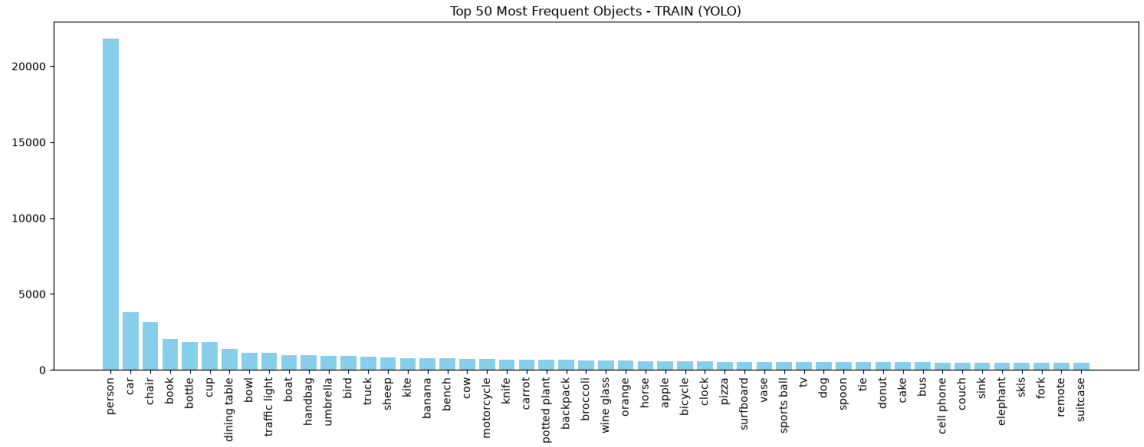


Figure 5: Top detected object categories in the Knowledge Base using YOLOv8.

4.3.2 Validation Results

YOLOv8 processed 100 images and detected 547 objects (5.47 per image).

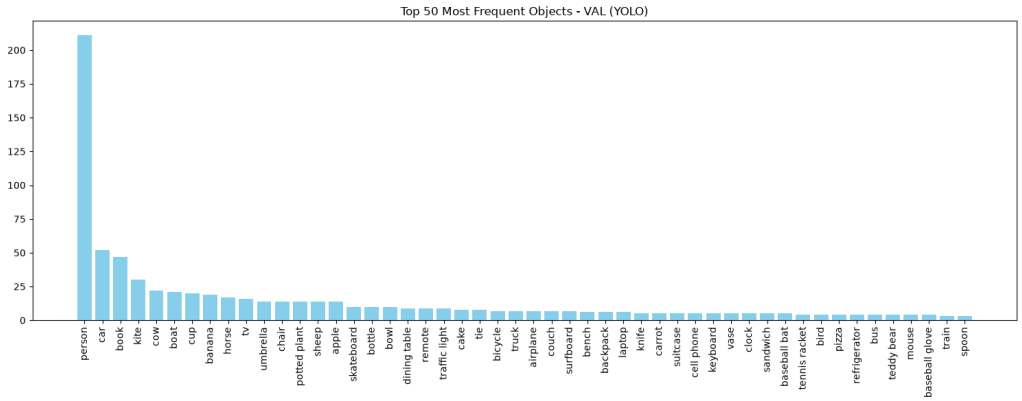


Figure 6: Top detected object categories in the Validation set using YOLOv8.

4.3.3 Test Results

YOLOv8 processed 1,000 images and detected 5,074 objects (5.07 per image).

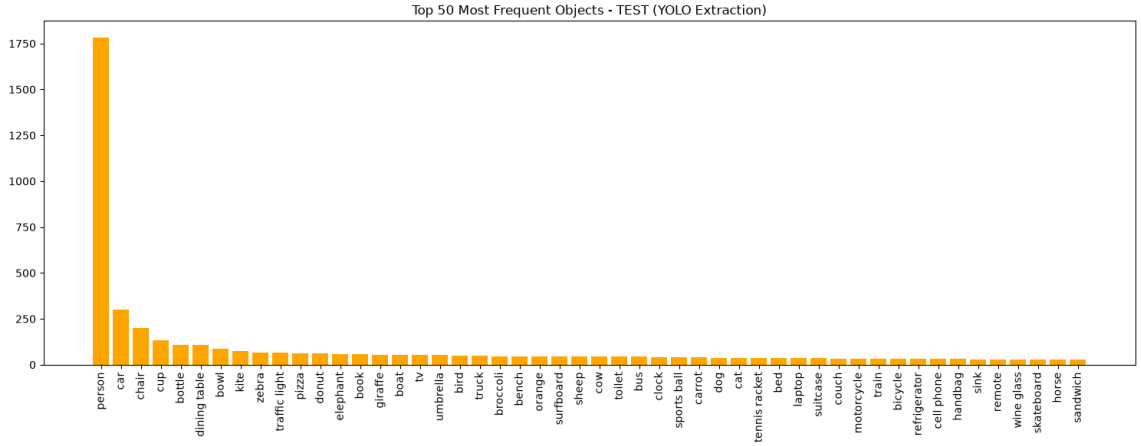


Figure 7: Top detected object categories in the Test set using YOLOv8.

4.4 Dataset Splitting and Anti-Leakage Audit

To prevent data leakage, we compared the image IDs of the three dataset splits. The verification confirmed that there are zero shared image IDs.

5 CLIP Embedding Generation

We use the pretrained CLIP model openai/clip-vit-base-patch32. CLIP maps images and text into a shared 512-dimensional vector space.

6 FAISS Vector Database

We created three FAISS indices using the Knowledge Base embeddings. We used IndexFlatIP, which performs exact inner-product similarity search.

7 Initial Retrieval Evaluation

We evaluated the initial CLIP and FAISS retrieval system using 200 held-out queries. The baseline results for Text-to-Image and Image-to-Text search are detailed in the full report.

8 Conclusion

In Phase 1, we prepared and explored the MS COCO dataset and established the initial retrieval foundation for our image captioning project. The dataset and retrieval components are now prepared for the next phase.